

Manual

Recording of Changes to Master Data SIS-ASD 4.0_{pe}

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1 Recording Changes in Master Data

Purpose

HYDRA logging is a basic function within the HYDRA MES Weaver that can be used to enter changes to a data object (master data/ configuration). Depending on the configuration, optional or mandatory comments can be entered concerning changes that have been made.

An evaluation can be used to illustrate changes that have been made. Depending on the type of recording, changes can be illustrated or different versions of an object compared to one another.

Implementation considerations

You employ this component if:

- You basically would like to keep track of changes made to an object
- You would like to have traceability for reasons of product or manufacturer documentation to meet legal regulations
- You would like to transfer changes to an object to connected systems via an interface

Integration

HYDRA logging represents a central function that is available for all master data objects.

Features

- Configuration:
 - Definition of whether changes to a data object (e.g. master data for machines, tools, staff) should be recorded.
 - Definition of whether a comment entry is optional or mandatory when a data object is newly created/ modified/ deleted
- Evaluation:
 - List of changes to a data object (old vs. new value)
- Version administration:
 - List of changes to a data object (old vs. new value)
 - Undo function used to re-activate an earlier version of the data object

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ASD_4.0\MBL_Base_system_logging.docx>>

2 Logging Configuration

Overview

HYDRA menu	System administration → Logging → Logging - configuration
FEDRA menu	System administration → Logging → Logging - configuration
Transaction code	logcfg
Function authorization	logcfg.*

Purpose

Use the application if you want to change or activate the storage for changes of objects.

Integration

The logging function is a central function available for all master data objects, dialogs and services.

Selection criteria

The application provides the following selection criteria:

Service

Logging of service calls.

Dialog

Logging of dialog calls for a called dialog.

Object

Logging of objects that are firmly stored in the software including the respective values. You can restrict logging to the actions of the respective objects. In this case, you cannot enter comments / labels.



If you generate logging configurations, you must also consider/carry out the archiving of the generated logging data.

Object logging

Application area

HLS-Save

Field descriptions

Object

The object identifies a system data object.

Example:

You want to log changes of the machine configuration. The object identification for the machine is "MNR". In this case, you must define "MNR" as object.

Action

The "action" field specifies how an object has been changed, i.e. whether it has been created (INSERT), changed (UPDATE) or deleted (DELETE).

Create an object: INSERT

Change an object: UPDATE

Delete an object: DELETE

You can use the placeholder "%" for an action if you want to log the changes made to an object irrespective of the method in use.

Example:

You want to log all changes made to the machine configuration. The object identification for the machine is "MNR". In this case, you must define "MNR" as object and "%" as action.

You only want to log changes (UPDATE) of the machine configuration. The object identification for the machine is "MNR". In this case, you must define "MNR" as object and "UPDATE" as action.

Log mode

You can specify the log mode for the object.

Key

In this case, the system only logs fields of the object that are configured as keys.

Call

The entire call is logged.

Signalization

To use the logging as source for uploads to external systems, you can use the "SIGUSR" communication to upload data "online". Here the feedback program is informed by the logging application. To let the system know which program is to be informed, enter the program name here in lower case letters without file extension, e.g. "myerprck".

Segment

When data is uploaded from the *system logging* function to external systems, the logging configuration must include a segment name. If you specify the segment name, the upload program can identify the data records that must be uploaded.

Log ERP data

This option activates the logging of data transferred via interfaces from external systems.

If the option is activated, interface data is logged as well. If the entry of a comment or label is enabled for the corresponding object and method, it will be ignored when interface data is logged.

Logging per dialog

Application area

- Terminal dialogs
- OP logons via MOC
- Interface transactions via BAPI

Field descriptions

Dialog

Definition of the BAPI call you want to log.

Log mode

You can specify the log mode for the object.

- Key
In this case, only fields configured as keys to the object are logged by the system.
- Call
The entire call is logged.
- Total data
The call is logged and data that is not included in the call is reloaded.

Comment

- Mandatory
You must enter a comment when you carry out the dialog. The input field for the comment is opened and the action cannot be performed if you do not enter a comment.
- Optional
The input of a comment is optional when you carry out the dialog. A dialog to enter a comment is opened, but you can also perform the action without including a comment.
- No entry
You cannot enter a comment when you carry out the dialog.

Labeling

- **Mandatory**
You must enter a label when you carry out the dialog. The input field for the label is opened and the action cannot be performed if you do not enter a label.
- **Optional**
The input of a label is optional when you carry out the dialog. A dialog to enter a label is opened, but you can also perform the action without including a label.
- **No entry**
You cannot enter a label when you carry out the dialog.

Signalization

To use the logging as source for uploads to external systems, you can use the "SIGUSR" communication to upload data "online". Here the feedback program is informed by the logging application. To let the system know which program is to be informed, enter the program name here in lower case letters without file extension, e.g. "myerprck".

Segment

When data is uploaded from the system logging function to external systems, the logging configuration must include a segment name. If you specify the segment name, the upload program can identify the data records that must be uploaded.

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This option activates the logging of data transferred via interfaces from external systems.

If the option is activated, interface data is logged as well. If the entry of a comment or label is enabled for the corresponding object and method, it will be ignored when interface data is logged.

Logging per service

Application area

- Service calls (clients/interface)

Field descriptions

Service

Definition of the service call you want to log.

Log mode

You can specify the log mode for the service call.

- Key

In this case, the system only logs fields of the service that are configured as keys.

- **Call**
The entire service call is logged.

Comment

- **Mandatory**
You must enter a comment when you carry out the service. The input field for the comment is opened and the service call cannot be performed if you do not enter a comment.
- **Optional**
The input of a comment is optional when you carry out the service. A dialog to enter a comment is opened, but you can also carry out the service call without including a comment.
- **No entry**
You cannot enter a comment when you carry out the service call.

Labeling

- **Mandatory**
You must enter a label when you carry out the service. The input field for the label is opened and the service call cannot be performed if you do not enter a label.
- **Optional**
The input of a label is optional when you carry out the service call. A dialog to enter a label is opened, but you can also carry out the service call without including a label.
- **No entry**
You cannot enter a label when you carry out the dialog.

Signalization

To use the logging as source for uploads to external systems, you can use the "SIGUSR" communication to upload data "online". In this context, the upload program is actively informed by the logging application. To let the system know which program is to be informed, enter the program name here in lower case letters without file extension, e.g. "myerprck".

Segment

When data is uploaded from the HYDRA logging function to external systems, the logging configuration must include a segment name. If you specify the segment name, the upload program can identify the data records that must be uploaded.

Key fields (key / key name)

You can use the specified key fields and their respective name to filter data within the evaluations.

Key 1

The domain of the service that you want to log is preset here.

Key name 1

The name of the domain (source repository) is preset here.

Key 2-5

The key fields of the service that you want to log are available for selection.

Key name 2-5

The respective key names of the keys of the service that you want to log.

Key word columns

You can assign key words to the configured service. These key words are available within the evaluation. You can assign each acronym of the configured service as key word.

3 Logging Key Maintenance

Usage

Logging keys need to be maintained, as the logging function does not have any information about the key fields of an object if the changed data or the entire data are to be logged. Only does the unique key allow tracing back the changes made to an object. In this context, the system generates a consecutive change order for each object/key field combination. These keys can only be edited by the application.

Integration

The system can record changes to objects on the basis of the information that is kept here.

Field description

Object

Just as it is the case for the logging configuration, the object is to be entered in this field. In this case, it does not play a role if only key fields, changes or the entire object is to be logged.

Example:

All changes to the machine configuration are to be logged. "MNR" is the object identification for logging changes.

Action

The action can be defined in this field. Since the key fields for most of the BAPIs are identical irrespective of the method, "%AKTION%" can be used as placeholder to be able to represent all actions by one entry.

Key 1

The key 1 of a BAPI corresponds to the object. This means that "MNR" is to be defined for the above example.

Designation 1

A designation may be defined for key 1. This name will be reused in evaluations/reports.

Key 2

Key 2 and the following keys identify a single object. Please refer to annex A for further information on which values may be defined here.

Designation 2

Designation 2 and the following designations are the keys to identify a single object. Please refer to annex A for further information on which values may be defined here.

Key 3

See key 2

Designation 3

See designation 2

Key 4

See key 2

Designation 4

See designation 2

Key 5

See key 2

Designation 5

See designation 2

4 Logging - Change Management

5.1 Summary

Menu	System administration → Logging → Logging - Change management
Transaction code	atrail
Function authorization	atrail.view - Display atrail.diff – Show differences

Purpose

You use Logging Change Management if you want to compare several recorded changes to one object.

Integration

Logging is a central functional feature available for all master data objects, dialogs and services.

Field descriptions

Modified on

Here you can specify the selection area to display logging entries.

Modified by

Modified by/ trigger of a logging entry.

Input type

- New

Identifies a recreated entry.

- Change

Identifies an existing, edited entry.

←←

Labeling

Documented labeling when recreating / changing / deleting.

Comment

Documented comment when recreating / changing / deleting.

Searching for keywords

The system filters keywords. Configuration / assignment of keywords during logging configuration.

Key 1

Key 1 of a BAPI corresponds to the object. In the example above, please store "MNR" (machine number).

Key 2

Key 2 and the following keys are used to identify a single object.

Key 3

Refer to key 2.

Key 4

Refer to key 2.

Key 5

Refer to key 2.

Toolbar**Display of logging data**

Display logging data of the logged dialog. Requirement to display dialog data is activation of the object "Logging dialog data" in the logging configuration of the object and maintaining the logging key.

Display differences

Display differences between 2 logged dialogs. You must beforehand mark the required versions for the compared dialogs. While pressing the STRG key when selecting the first entry, please select the second entry. Then select the function you want to compare.

Requirement to display dialog data is activation of the object "Logging dialog data" in the logging configuration of the object and maintaining the logging key.

5 Logging-Specific Configurations

Overview

Menu	System administration → Archiving → Data management
Transaction code	arccfg
Function authorization	arccfg.*

By default, the logging data is stored in the online data set for 35 days. After this time, the data is moved to archive tables. In the archive tables, the retention period for logging data is three years. After this time, the data is deleted.

Configuration

Configure the archiving in the application *Data management*. The archiving function provides the following entries:

Parameter	Value
Product	AUDITLOG
Object	LOG
Retention period	35
Retention period in	Day
Action	Archiving
Condition	(key1 <> "HLS" or key1 is null)
Administration table	arc_verw_hyd
Administration duration	10
Administration duration in	Year
Configuration	Yes
Comment	Archiving AuditLog
Archiving type	Z

Priority	100
Master table	hyd_logging
Date column	datum
Key 1	verweis

Parameter	Value
Product	AUDITLOG
Object	A_LOG
Retention period	3
Retention period in	Year
Action	Delete
Condition	
Administration table	arc_verw_hyd
Administration duration	10
Administration duration in	Year
Configuration	Yes
Comment	Archiving AuditLog
Archiving type	Z
Priority	100
Master table	a_hyd_logging
Date column	datum
Key 1	verweis



When the data has been deleted from the archive tables, you cannot reload this data any more.
Reason: The logging data is archived in BLOB fields and is deleted from the archive tables when the retention period has expired.

Version upgrade to MW 4.0pe

After the version upgrade to MW4.0pe, you should use the current logging archiving after consultation with the customer.

The patch dbp_archive_auditlog.hsc adds the above-mentioned configurations to the data management.

Windows

hydscr.exe db_sql\dbp_archive_auditlog.hsc

Linux

hydscr.out db_sql/dbp_archive_auditlog.hsc

Check if the following entry is available in the archiving script hyarc.scr (d:\mip1):

```
# Start AuditLog archiving
if [ `hyd_patchchk.exe -r ARCHIVE_AUDITLOG` -gt 0 ]
then
  echo "AuditLog:" >> $ERRPATH/hyarc.pro
  hymwarc.exe -d -z "PROD=AUDITLOG|REORG_VERW_TABLE=1|" >
  $ERRPATH/hymwarc_auditlog.pro
  cat $ERRPATH/hymwarc_auditlog.pro >> $ERRPATH/hyarc.pro
  sleep 1
fi
# End AuditLog archiving
```

If required, you can copy the entry from the template hyarc.bsp.

6 Logging Key Overview

Summary

Please note:

The "action" field is used as "%AKTION%" in any case. For lack of space, this is not stated in each individual case.

USRFLDDE F	USRFLDCF G	MSTTXT	AUART	NRKREIS	BEARBFKT	FKTPROF	MNR	OBJEKT
USRFLDDE F	USRFLDCF G	MSTTXT	AUART	NRKREIS	BEARBFKT	FKTPROF	MNR	KEY1
%USRFLDD EF.KENN%	%USRFLDC FG.OBJTYP %	%MSTTXT. STNR%	%AUART.A UART%	%NRKREIS. OBJTYP%	%BAERBFK T.BEARB%	%FKTPROF .FKTPROF %	%MNR.MN R%	KEY2
%USRFLDD EF.KENN:Z %	%USRFLDC FG.USRFLD %			%NRKREIS. ART%	%BEARBFK T.ART%	%FKTPROF .FKT%		KEY3
					%BEARBFK T.FKT%	%FKTPROF .BERECHT %		KEY4
					%BEARBFK T.BERECHT %			KEY5
field type definitions	user field key	status texts	order type	number range	function authorizatio ns	function profiles	machines	KEYNAME1
field type	object type	status number	order type	objecttype	user	function profil	machinnun ber	KEYNAME2
	userfield			form	type	function		KEYNAME3
					function	authorizatio n		KEYNAME4
					authorizatio n			KEYNAME5

RESWART	STKL	BDEJMOD	MDEFEITG	BDETMOD	GRPRES	MNRTNR	GR	GRTXT	USRFLDEL EM	OBJEKT
RESWART	STKL	BDEJMOD	MDEFEITG	BDETMOD	GRPRES	MNRTNR	GR	GRTXT	USRFLDEL EM	KEY1
%RESWART: T. VERWEIS %	%STKL. STK L%	%BDEJMO D. JAHR%	%MDEFEIT G. DAT%	%BDETMO D. SKMODN R%	%GRPRES. GRP%	%MNRTNR. MNR%	%GR.MNR %	%GRTXT.G RTXTNR%	%USRFLDE LEM.OBJTY P%	KEY2
	%STKL. STK LNR%		%MDEFEIT G. VERWEIS %		%GRPRES. RESTYP%	%MNRTNR. TNR%	%GR.ART%		%USRFLDE LEM.USRFL D%	KEY3
			%MDEFEIT G. JAHR%		%GRPRES. RES%		%GR.GR%		%USRFLDE LEM.IDX%	KEY4
										KEY5
maintenanc e calndar	Disturbance classes	year model	pub.holidays	day types	group assignment	machine- terminal assignment	reason assignment	reason text	user fields	KEYNAME1
reference		year	date	day type	group	machine	workplace	reason text	object type	KEYNAME2
			reference		resourcety p e	terminal	form		user field keys	KEYNAME3
			year		resource		reason		field-ID	KEYNAME4
										KEYNAME5

BMK	LIC	EXPR	EINHUMR	VACODE	EINH	BPOS	DLGCFG	TGRP	ZYP	OBJEKT
BMK	LIC	EXPR	EINHUMR	VACODE	EINH	BPOS	DLGCFG	TGRP	ZYP	KEY1
%BMK.BMK NR%	%LIC.CAT%	%EXPR.EX PR%	%EINHUMR .EINH.V%	%VACODE. VERARBCO DE%	%EINH.EIN H%	%BPOS.MN R%	%DLGCFG. VERWEIS%	%TGRP.TG RP%	%ZYP.MNR %	KEY2
	%LIC.PROK EY%		%EINHUMR .EINH.Z%			%BPOS.BP OS%	%DLGCFG. DLG%		%ZYP.MNR: Z%	KEY3
			%EINHUMR .MAT%				%DLGCFG. TYP%			KEY4
			%EINHUMR .MATYP%				%DLGCFG. DLGUSR%			KEY5
resource performance account	licenses	formula managemen t	unit conversion	processing codes	units		dynamic dialogs	terminal groups	cycle parameters	KEYNAME1
resourcenu mber	category	formula	unit from		unit		reference	terminal group	machine number	KEYNAME2
	product		unit to				dialog			KEYNAME3
			material				type			KEYNAME4
			material aplication type				user			KEYNAME5

BEARBPRO F	ASTTXT	AST	BEARBCFG	BEARB	BEARBVAB PROF	BEARBKST	PRNLAYCF G	INI	MLEBASCF G	OBJEKT
BEARBPRO F	ASTTXT	AST	BEARBCFG	BEARB	BEARBVAB PROF	BEARBKST	PRNLAYCF G	INI	MLEBASCF G	KEY1
%BEARBPR OF.BEARB %	%ASTTXT.A STTXTNR%	%AST.AUA RT%	%BEARBCF G.BEARB%	%BEARB.B EARB%	%BEARBVAB BPROF.VE RWEIS%	%BEARBKS T.BEARB%	%PRNLAYC FG.ID%	%INI.INI%	%MLEBASC FG.VERWEI S%	KEY2
%BEARBPR OF.BEARB PROF%		%AST.ASTT YP%				%BEARBKS T.FIR%	%PRNLAYC FG.TYP%	%INI.USR%	%MLEBASC FG.MESTY P%	KEY3
%BEARBPR OF.VERWEI S%		%AST.AST %				%BEARBKS T.KST%			%MLEBASC FG.MESVA R%	KEY4
									%MLEBASC FG.MESFC T%	KEY5
	status text	order status	account rules	user administratio n	responsibilit y areas	cost center authorizatio n	report configuratio n	INI configuratio n	MLE basic configuratio n	KEYNAME1
	status textnumber	order type	user	user	reference	uer	configuratio n	INI	reference	KEYNAME2
		status type				company	type	user	message typ e	KEYNAME3
		status				cost center			variant	KEYNAME4
									function	KEYNAME5

TNR	STATXT	PNR	MST	MATMATP	MATTYPTE	MATTYP	MATPUF	LSTCODE	GRP	OBJEKT
TNR	STATXT	PNR	MST	MATMATP	MATTYPTE	MATTYP	MATPUF	LSTCODE	GRP	KEY1
%TNR.TNR %	%STATXT.S TATXTNR%	%PNR.PNR %	%MST.MNR %	%MATMAT TYP.MAT%	%MATTYPTE PE.MATTYP %	%MATTYP. MATTYP%	%MATPUF. MATPUF%	%LSTCODE .LSTCODE %	%GRP.GRP %	KEY2
	%STATXT.A TYP%		%MST.MST %		%MATTYPTE PE.TPE%			%LSTCODE .LST%		KEY3
										KEY4
										KEY5
terminals		HR master data	machine status			material types	material buffer	automatic status change	groups	KEYNAME1
terminalnum ber		personnel number	machine	material	materialtyp	material type	material		group	KEYNAME2
			status							KEYNAME3
										KEYNAME4
										KEYNAME5

PLANVAR	BEARBPRO F	MATLIST	FERTVAR	RUEW	MST	MATMATPU F	WART	VABPROF	TPE	OBJEKT
PLANVAR	BEARBPRO F	MATLIST	FERTVAR	RUEW	MST	MATMATPU F	WART	VABPROF	TPE	KEY1
%PLANVAR .PLANVAR %	%BEARBPR OF.BEARB %	%MATLIST. ANR%	%FERTVAR .VERWEIS %	%RUEW.TY P%	%MST.MNR %	%MATMAT PUF.MAT%	%WART.MN R%	%VABPROF .VABPROF %	%TPE.TPE %	KEY2
	%BEARBPR OF.BEARB PROF%			%RUEW.M GRP%	%MST.MST %	%MATMAT PUF.MATP UF%				KEY3
	%BEARBPR OF.KENN%									KEY4
	%BEARBPR OF.VAL%									KEY5
								responsibilit y profiles	transport unit	KEYNAME1
						material				KEYNAME2
						materialbuff er				KEYNAME3
										KEYNAME4
										KEYNAME5

RESSTA	RESFAM	RESTYP	RES	OBJEKT
RESSTA	RESFAM	RESTYP	RES	KEY1
%RESSTA. RESTYPID %	%RESFAM. RESFAM%	%RESTYP. RESTYP%	%RES.RES TYP%	KEY2
		%RESTYP. RESTYEXT %	%RES.RES %	KEY3
		%RESTYP. PLAUS:BEL %		KEY4
				KEY5
				KEYNAME1
				KEYNAME2
				KEYNAME3
				KEYNAME4
				KEYNAME5